SELECTION OF NUMERICAL TECHNIQUES

Field zone	Topology	Evaluated quantity	Suitable numerical technique
Near-field	Open	Field	FDTD, MOM
Near-field	Open	SAR	FDTD
Near-field	Closed, multiple scatterers	Field	FDTD, MOM
Near-field	Closed, multiple scatterers	SAR	FDTD, MR/FDTD
Far-field	Open	Field	Ray tracing, MOM
Far-field	Multiple scatterers (complex urban environment)	Field	Ray tracing

base station antenna. EIRP is the product of power supplied to the antenna and maximum antenna gain relative to an isotropic antenna. In this method, site survey is done and several parameters like location parameters, operating parameters and environment parameters are recorded.

Assessment of EIRP and threshold of EIRP ($EIRP_{th}$) are done at various publicly accessible points (on ground, rooftop, adjacent building, etc) in the environment surrounding the base station antenna.

Threshold value of EIRP should be calculated at the position within the exposure assessment area where power density is maximum. The most important parameter for determining exposure due to elevated antennae (broad coverage antenna—omni directional or sectional) is the vertical (elevation) antenna pattern. Since exposure assessment assumes exposure along the direction of maximum radiation in the horizontal plane, horizontal (azimuth) pattern is not relevant.

For a shared site, EIRP and $EIRP_{th}$ have be to calculated. Cumulative ratio should be less than unity at all points outside the exceedance zone for normally compliant site.

Calculation facts. The beauty of exposure assessment by calculation is to pre-estimate the exposure from non-existent radiation sources. Calculations consider maximum possible radiation (ERP) that leads to maximum possible exposure levels. It is also possible to make calculations in areas with no access. It gives an opportunity to apply mitigation techniques to reduce radiation levels, if required.

Accuracy of results depends on the model used and on the exactness of description of transmitting antennae. Influence of reflections is ignored in the calculations that may degrade the accuracy. For very accurate results, a detailed and precise description of the radiating antenna and the exposure environment is required.

Conclusion

There is a strong perception related to the existence of a high level of EM radiation in the vicinity of telecom towers, which may cause adverse biological effects. So, it is about time an exhaustive assessment of radiation levels and their possible health effects is made, and guidelines for safety are laid down accordingly. If EM radiation levels in any area accessible to humans are higher than the prescribed limits, it is strongly recommended to take necessary action to reduce the radiation levels. EFY